

Md. Abdur Rahman

AI Engineer · LLM Systems · RAG & Fine-Tuning · Multi-Agent AI

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Summary

AI Engineer building production LLM systems: retrieval pipelines, multi-agent orchestration, and model fine-tuning. Selected as **Betopia AI Champion**, ranked the top AI Engineer across all of Betopia's business units (2026). I focus on self-hosted LLM infrastructure that runs at enterprise scale on-premises, using vLLM for GPU serving, LangGraph for agent workflows, RAG, and LoRA/QLoRA/GRPO for fine-tuning, in Python and FastAPI.

Experience

Betopia Limited

AI Engineer — company-wide AI Champion

Mar 2026 – Present

On-site, Dhaka, Bangladesh

- Built **Betopia.ai** ecosystem, the company's internal AI platform: LLM API gateway, web chat, a coding-agent CLI, and an IDE extension. In beta with **700+** employees and used daily by the engineering team.
- Self-hosted **Qwen3.6**, **Ornith-1.5** on **vLLM** on one 96 GB NVIDIA Blackwell GPU, removing the third-party inference bill entirely. Held time-to-first-token under a second at peak concurrency using continuous batching and prefix caching.
- Built the agent runtime on **LangGraph** — multi-turn state, tool calling, **Agentic Search Engine**, sandboxed code execution, repo-aware retrieval — streamed to clients over SSE.
- Wrote the context layer that assembles each prompt from system, project, repository, and session sources, with summarization and eviction to keep long sessions inside the context window.

SM Technology

AI Engineer — a Strategic Business Unit of Betopia Limited

Nov 2025 – Feb 2026

On-site, Dhaka, Bangladesh

- Designed **Firecomm AI**, a role-aware **Agentic RAG platform** for **2,200+** enterprise users: RBAC-scoped retrieval, hybrid search (dense vector + BM25), and cited answers over 500+ operational documents.
- Built an **AI fitness platform** with LLM-generated workout and nutrition plans, social features, and a multi-vendor marketplace, on a modular **FastAPI** backend containerized with Docker.
- Implemented a four-stage **recommendation pipeline** — retrieval, ranking, cross-encoder re-ranking, metadata filtering — on **PGVector** with dense embeddings, improving product-discovery relevance for e-commerce.
- Shipped a **voice-enabled RAG** system over a large educational corpus, using hybrid dense/sparse retrieval and tuned inference to keep responses fast and grounded.

Projects

Lawyer.AI — Legal-Domain Multi-Agent Chatbot 🤖

Python · RAG · LangChain · LangGraph · Hugging Face · ChromaDB · Streamlit

Aug 2025

- Built a legal-domain chatbot using a LangGraph multi-agent workflow for context-aware document retrieval and step-by-step legal reasoning.
- Implemented a hybrid RAG pipeline combining dense (embedding) and sparse (BM25) retrieval with cross-encoder re-ranking, measurably improving answer relevance over a single-retrieval baseline.
- Added citation-aware prompting and SQLite conversation checkpointing for fully traceable, multi-turn consultations.

GRPO Reinforcement-Learning Fine-Tuning 🤖

Python · Hugging Face TRL · GRPO · RLHF · Transformers · Jupyter

Jan 2026

- Applied Group Relative Policy Optimisation (GRPO) to align model outputs with reward signals, going beyond supervised fine-tuning toward human-preference alignment.
- Implemented a full RLHF training loop in Hugging Face TRL, including reward modelling and policy-gradient optimisation, to improve instruction-following.

Bangla Hate-Speech Classifier — NLP Fine-Tuning 🤖

Python · Hugging Face · PEFT · LoRA · QLoRA · BanglaBERT

Feb 2025

- Fine-tuned BanglaBERT with LoRA/QLoRA (PEFT) adapters for multi-class hate-speech detection, reaching **72% accuracy** through targeted data balancing and text normalisation.
- Reduced the GPU memory footprint by **≈60%** via 4-bit quantisation (QLoRA), enabling fine-tuning on consumer hardware with no measurable accuracy loss.

Expense Tracker — MCP Server 🤖

PostgreSQL · FastMCP · Claude AI · Model Context Protocol (MCP)

Oct 2025

- Built a natural-language personal-finance tracker that bridges Claude tool-use to a PostgreSQL backend over the Model Context Protocol (MCP) for multi-user management.

Additional Projects

- **Email Classification System** – FastAPI, scikit-learn, React; **92% accuracy** on spam/ham using TF-IDF and ensemble classifiers.
- **AI Search Agent**  – Hugging Face LLMs, Tavily API, Gradio; multi-session memory for stateful, web-augmented question answering.
- **EduConnect** – full-stack education marketplace (React, Node.js, MongoDB, Firebase) with geolocation tutor search and real-time notifications over Socket.IO.

Technical Skills

AI / ML: LLMs, Agentic Chat, RAG, multi-agent systems, fine-tuning (LoRA, QLoRA, GRPO, RLHF), PEFT, transformers, NLP, vector search, deep learning

LLM Serving & APIs: vLLM (GPU inference), Hugging Face Transformers, TRL, OpenAI API, Anthropic Claude API, Model Context Protocol (MCP)

Frameworks & Orchestration: LangChain, LangGraph, FastAPI, Streamlit, Gradio, Node.js, Express.js, React

Data & Vector Stores: PostgreSQL, PGVector, ChromaDB, MongoDB, Redis

Languages: Python, SQL, JavaScript, Java, C

Tools & MLOps: Docker, Docker Compose, MLflow, Git, Pandas, NumPy

Education

Green University of Bangladesh
B.Sc. in Computer Science and Engineering

Jan 2022 – Feb 2026

Publications

Md. A. Rahman*, M. N. Hasan, M. S. Hossain, W. Tasnim, and S. S. Nakshi, "Enhanced Hybrid Deep Learning Approach for Heart Attack Prediction Using ECG Data," presented at the 5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026), Natore, Bangladesh, Aug. 2026.

Achievements

- **Betopia AI Champion (2026)** – ranked the top AI Engineer across all Strategic Business Units of Betopia Limited in a company-wide competitive selection.
- Ranked among the top participants at the **BLP Workshop** and **BUBT AI Conquest** competitive AI/ML events.
- Solved **100+ LeetCode problems**, focused on dynamic programming, graph algorithms, and system-level optimisation.
- Led multiple university software projects using Agile/Scrum: sprint planning, code review, and cross-functional coordination.

Languages

English (Fluent) **Bengali** (Native) **Hindi** (Fluent)